Uni:

Section 1: True or false? Circle the appropriate choice (1 point each).

1. (TRUE / FALSE) In a simple linear regression model, the expected value of Y when X has the value x^* is $\beta_0 + \beta_1 x^*$.

2. (TRUE / FALSE) The value of the correlation between two variables depends on which variable is labeled X and which is labeled Y.

3. (TRUE / FALSE) If the data $\{(x_1, y_1), \ldots, (x_n, y_n)\}$ are a random sample from a population for which the simple linear regression model is true, the least squares estimator of the slope provides an unbiased estimate of the population slope β_1, meaning there is no systematic tendency for $\hat{\beta}_1$ to be too high or too low.

4. (TRUE / FALSE) For a simple linear regression model, if a hypothesis test fails to reject the null hypothesis that the slope equals zero, you can conclude there is no relationship between X and Y.

5. (TRUE / FALSE) For any simple linear regression model we fit to data, the sum of squared residuals $\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$ will always be less than or equal to the total sum of squares $\sum_{i=1}^{n}(y_i - \bar{y})^2$.

6. (TRUE / FALSE) To check the linear regression assumption that the error terms have an expected value of zero, we can compute the mean of the residuals. If the mean of the residuals is not close to zero, this indicates a violation of the model assumptions.

7. (TRUE / FALSE) We are interested in estimating μ, the true average amount of sleep per night among adults in the US. Based on a random sample, we obtain a 95% confidence interval of 6.8 to 7.3 hours. Thus, we can conclude there is a 95% probability that the true average sleep duration is between 6.8 and 7.3 hours.

8. (TRUE / FALSE) One advantage of linear regression is that it allows us to make reliable predictions for Y at X values beyond the range in the observed data.

9. (TRUE / FALSE) When estimating a population proportion p, confidence interval widths are based on the t-distribution rather than the standard normal distribution, to account for the extra uncertainty that comes from not knowing the true standard deviation.

10. (TRUE / FALSE) The Central Limit Theorem provides a basis for statistical inference by guaranteeing that, in sufficiently large random samples, the sample values approximately follow a normal distribution.

Section 2: Circle the appropriate choice (2 points each).

11. Suppose X and Y are random variables, and that the variance of X is 16, and the variance of Y is 9. What is the standard deviation of X-Y?

    (a) 7

    (b) 5

    (c) 6.5

    (d) 25

    (e) cannot be determined with the information given

12. Suppose an internet service provider reports that 30% of its customers are dissatisfied, but we believe the true proportion is higher, and we want to test this. We sample 200 customers, and 40% of them report being dissatisfied. What are the appropriate null and alternative hypotheses for our question?

    (a) $H_0 : p = 0.4,\ H_a : p > 0.4$

    (b) $H_0 : \hat{p} = 0.4,\ H_a : \hat{p} > 0.4$

    (c) $H_0 : \hat{p} = 0.3,\ H_a : \hat{p} > 0.3$

    (d) $H_0 : p = 0.3,\ H_a : p > 0.3$

13. Suppose a hypothesis test yields a p-value of 0.03. Which of the following is the best interpretation?

    (a) There is a 3% chance that the null hypothesis is true.

    (b) There is a 3% chance of observing data this extreme or more extreme if the null hypothesis is true.

    (c) There is a 97% chance the alternative hypothesis is true.

    (d) The alternative hypothesis is 97% more likely to be true than the null hypothesis.

14. An experimenter performed a hypothesis test with a two-sided alternative hypothesis on a set of data and obtained a p-value of 0.44. For the same data, if the experimenter used a one-sided alternative instead, which of the following is true about the possible p-value(s) that the experimenter could have obtained?

    (a) the only possible p-value is 0.22

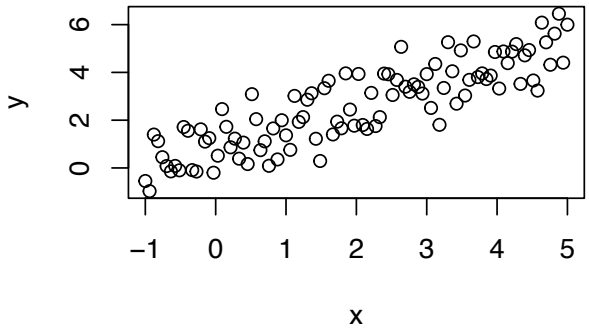
    (b) the only possible p-value is 0.44

    (c) the only possible p-value is 0.88

    (d) the possible p-values are 0.22 and 0.78

    (e) the possible p-values are 0.22 and 0.88

15. Suppose we fit a linear regression model to the data depicted in the scatter plot below, using X as the predictor and Y as the response, and obtain an estimated slope of $\hat{\beta}_1$.



If we reversed the variables, fitting a linear regression model using Y as the predictor and X as the response, the new estimated slope would be:

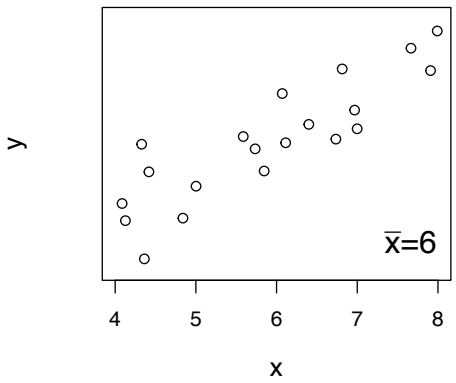
(a) equal to $\frac{1}{\hat{\beta}_1}$

(b) less than $\frac{1}{\hat{\beta}_1}$

(c) greater than $\frac{1}{\hat{\beta}_1}$

(d) cannot be determined with the information given

Questions 16-17 are related to the simple linear regression model fit to data depicted in the scatter plot below.



16. Which of the following will be wider:

(a) 99% confidence interval for the average Y value when X=5

(b) 99% prediction interval for an individual Y value when X=5

(c) cannot be determined with the information given

17. Which of the following will be wider:

    (a) 95% confidence interval for the average Y value when X=11

    (b) 95% prediction interval for an individual Y value when X=6

    (c) cannot be determined with the information given

Section 3: Answer all questions in the space provided.

1. Central City Power computes the monthly cost of electricity for a customer as:

$$\text{Cost} = 0.12^*\text{usage} + 15,$$

   where cost is measured in dollars, and usage is measured in kilowatt-hours. The average usage each month is 1000, with a standard deviation of 50. What are the mean and standard deviation of the monthly cost?

$$\begin{aligned}\mathbb{E}(cost) &= \mathbb{E}(0.12 * usage + 15)\\ &= 0.12 * \mathbb{E}(usage) + 15\\ &= 120 + 15\\ &= \$135\end{aligned}$$

$$\begin{aligned}\text{var}(cost) &= \text{var}(0.12 * usage + 15)\\ &= (0.12^2) * \text{var}(usage)\\ \text{sd}(cost) &= \sqrt{(0.12^2) * \text{var}(usage)}\\ &= 0.12 * \text{sd}(usage)\\ &= 0.12 * 50\\ &= \$6\end{aligned}$$

2. An animal shelter finds that small dogs tend to be adopted more quickly than large dogs. Fitting a linear regression model using weight to predict days to adoption, they obtain a significant positive slope. The shelter director concludes that they should prioritize helping dogs lose weight in order to shorten their time to adoption. From a statistical standpoint, is this a reasonable conclusion? Explain why or why not.

   No. Although the significant slope confirms that there is a correlation between weight and adoption time, an overall trend where smaller dogs get adopted faster does *not* mean that for any individual dog, a decrease in weight would cause a shorter adoption time.

You might also point out contextual reasons why this doesn't make sense. For instance, a dog being "large" doesn't necessarily mean it is overweight, so weight loss might negatively impact their health, which would probably make it *harder* for them to get adopted, rather than easier.

3. Suppose that the height of mature pine trees is normally distributed, with standard deviation of 20 feet. In a random sample of 10 trees, heights had a sample average of 110 feet and sample standard deviation of 14. Give a 90% confidence interval for the true average height of pine trees.

   Population standard deviation is known, so an interval based on the normal distribution (not the t-distribution) is the appropriate choice.

   95th percentile of Normal distribution ≈ 1.64

$$\bar{x} \pm 1.64\sqrt{\frac{20^2}{10}}$$
$$110 \pm 10.4$$

   We are 90% confident that the true average height is between 99.6 and 120.4 ft.

4. The output below shows the results of fitting a linear regression model to predict zinc concentration (in parts per million, "ppm") from lead concentration (in ppm) in fish from a certain river. The analysis is based on data from 80 fish.

```
lm(formula = zinc_ppm ~ lead_ppm)
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)    16.3     8.62       1.89    0.059
lead_ppm       19.0     3.67       5.18    0.000
R-squared:  0.82
```

   (a) What zinc concentration does the model predict for a fish with a lead concentration of 0.8 ppm?

   $19.0 * 0.8 + 16.3 = 31.5$

   31.5 ppm

   (b) What is the residual for an observed fish with a lead concentration of 2 ppm and zinc concentration of 52 ppm?

$$\begin{aligned} \hat{y}_i &= 16.3 + 19.0 * 2 \\ &= 54.3 \\ y_i - \hat{y}_i &= 52 - 54.3 \\ &= -2.3 \end{aligned}$$

   The residual is -2.3 ppm.

(c) Find a 99% confidence interval for the slope of the population regression line. Describe the interpretation of the confidence interval.

$$19 \pm 2.58 * 3.67$$
$$19 \pm 9.47$$

A 99% confidence interval for the slope is approximately 9.5 to 28.5.

Here are some good and not-so-good examples of how you could describe the interpretation.

Good:

- for fish whose lead concentration differs by 1 ppm, we are 99% confident the true average difference in zinc concentration is between 9.5 and 28.5 ppm
- we are 99% confident that a 1 ppm difference in lead concentration is associated with a 9.5 to 28.5 ppm difference in average zinc concentration
- We are 99% confident that the true average difference in zinc concentration between fish with lead concentrations that differ by 1 ppm is between 9.5 and 28.5.

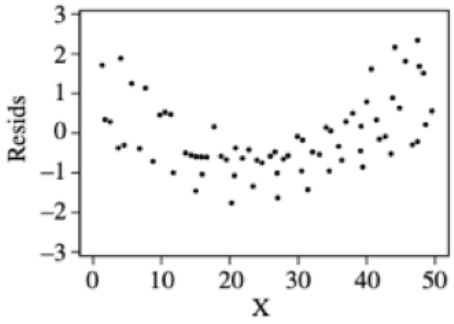
Not-so-good:

- we are 99% confident that when lead concentration increases by 1 ppm, zinc concentration increases by 9.5 to 28.5 ppm
- we are 99% confident that a 1 ppm increase in lead concentration is associated with a 9.5 to 28.5 ppm increase in zinc concentration, on average

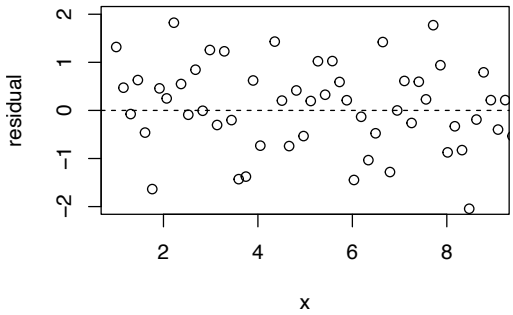
(d) Describe the interpretation of the R^2 value given above.

82% of the variation in observed zinc concentrations is explained by the fitted model.

5. (8 points) The plots below show the residuals from two different fitted simple linear regression models.



Plot 1



Plot 2

(a) In plot 1, which regression assumptions appear to be violated, if any?

Linearity. Appears that true relationship between x and y is non-linear.

(b) For the violated assumptions identified above (if any), what are the consequences for estimation of the linear model coefficients, and for inference (confidence intervals and hypothesis tests) based on the model?

If the true relationship is non-linear, the least squares estimators of slope and intercept are not correct (because there aren't corresponding true values of β_0 or β_1)

Inference based on the model is also incorrect.

(The question specifically asked about inference and the slope and intercept estimators, but it's also true that predictions made by the model can be arbitrarily poor.)

(c) In plot 2, which regression assumptions appear to be violated, if any?

None

(d) For the violated assumptions identified above (if any), what are the consequences for estimation of the linear model coefficients, and for inference based on the model?

N/A, no violated assumptions.